BATUAN NATIONAL HIGH SCHOOL

SUPREME SECONDARY LEARNER GOVERNMENT (SSLG) AUTOMATED VOTING & MANAGEMENT SYSTEM

Comprehensive System Architecture & Flowchart Documentation

System Name:	Batuan National High School — SSLG Voting System (batuan-voting)
Architecture:	React 18 (Vite SPA) + Node.js Express REST API + MySQL / Supabase
Target Roles:	Student Voters (Grades 7–12), SSLG COMELEC, System Administrators
Document Version:	Version 1.0 (Official Technical Specification & Customizable Appendix)

1. System Overview and Flowchart Architecture

The Batuan National High School SSLG Voting System is an automated web-based election and voter governance platform. The system provides complete end-to-end election workflows including voter credential management, candidate profiling, secure real-time ballot casting with atomic transaction safeguards, grade-level representation filtering, live tally broadcasting via WebSockets, official tally sheet generation, and past election archiving.

This document presents the complete system flowcharts structured in full accordance with the academic reference standard (matching the layout and decomposed off-page connector architecture of *CargoExpressFlowchart.docx*). Each figure is accompanied by an in-depth breakdown of user decisions, state transitions, validation trappings, and database mutations.

2. Flowchart Conventions and Symbol Key

Flowchart Shape	ANSI / ISO Name	Functional Role in Batuan Voting System
Oval / Stadium	Terminal Node	Represents the absolute system entry (START) and session termination (END).
Parallelogram	Input / Choices	Represents user interface menu choices, form input data (e.g. LRN, Password), and filters.
Diamond	Decision Gate	Evaluates boolean conditions (e.g. credentials valid, role=='admin', must_change_password, ongoing).
Rectangle	Process / Action	Represents password hashing, SQL database transactions, CSV parsing, and state updates.
Rounded Bullet	Display Output	Represents user-facing rendered UI screens, toast alerts, statistical summaries, and print dialogs.

3. Public Landing Page and Authentication Module

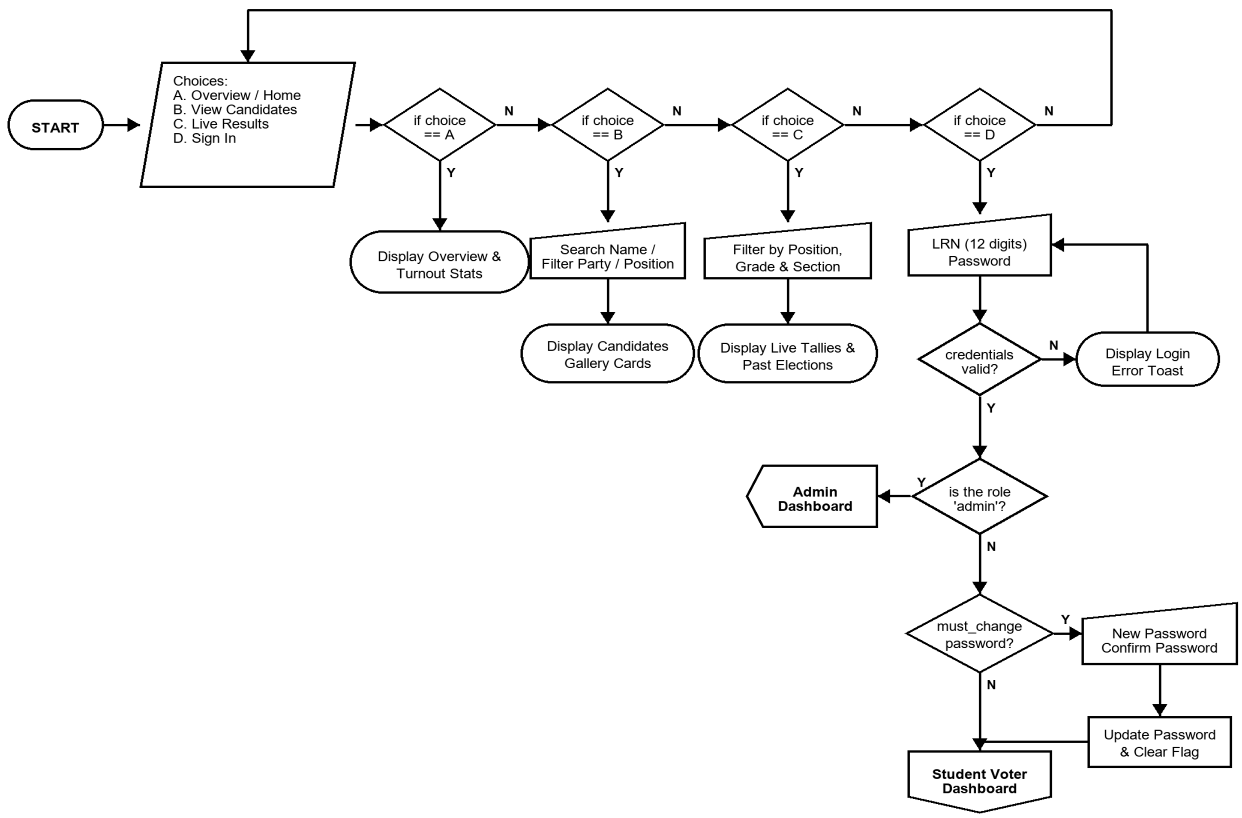

Figure 2. Landing Page & Authentication Flowchart

Figure 2 illustrates the entry gateway and public navigation for the Batuan Voting System. Upon loading the system at the root URL (/), users are presented with four primary choices (A through D):

- **Choice A (Overview / Home):** Displays the election hero banner, countdown timer, real-time voter turnout percentage, total registered voters, and the top 5 leading candidates leaderboard.
- **Choice B (View Candidates):** Provides a public candidate directory where students can search candidates by full name or party list, and filter by position.
- **Choice C (Live Results):** Renders the live election analytics tab with position, voter grade level, and section drilldowns, as well as archived past elections.
- **Choice D (Sign In Gate):** Accepts 12-digit Learner Reference Number (LRN) and password credentials. Submits to POST /api/auth/login.

Authentication Logic Trapping:
1. *Credentials Validation:* If credentials are invalid, an error toast is rendered and user loops back.
2. *Role Check:* If user role is 'admin', user is immediately routed to the Admin Dashboard (Connector 4).
3. *Forced Password Change:* If user is a student voter with must_change_password == true (e.g. newly imported voter whose default password equals their LRN), the system forces a redirect to /change-password before granting access.

4. Student Voter Portal and Ballot Casting Module

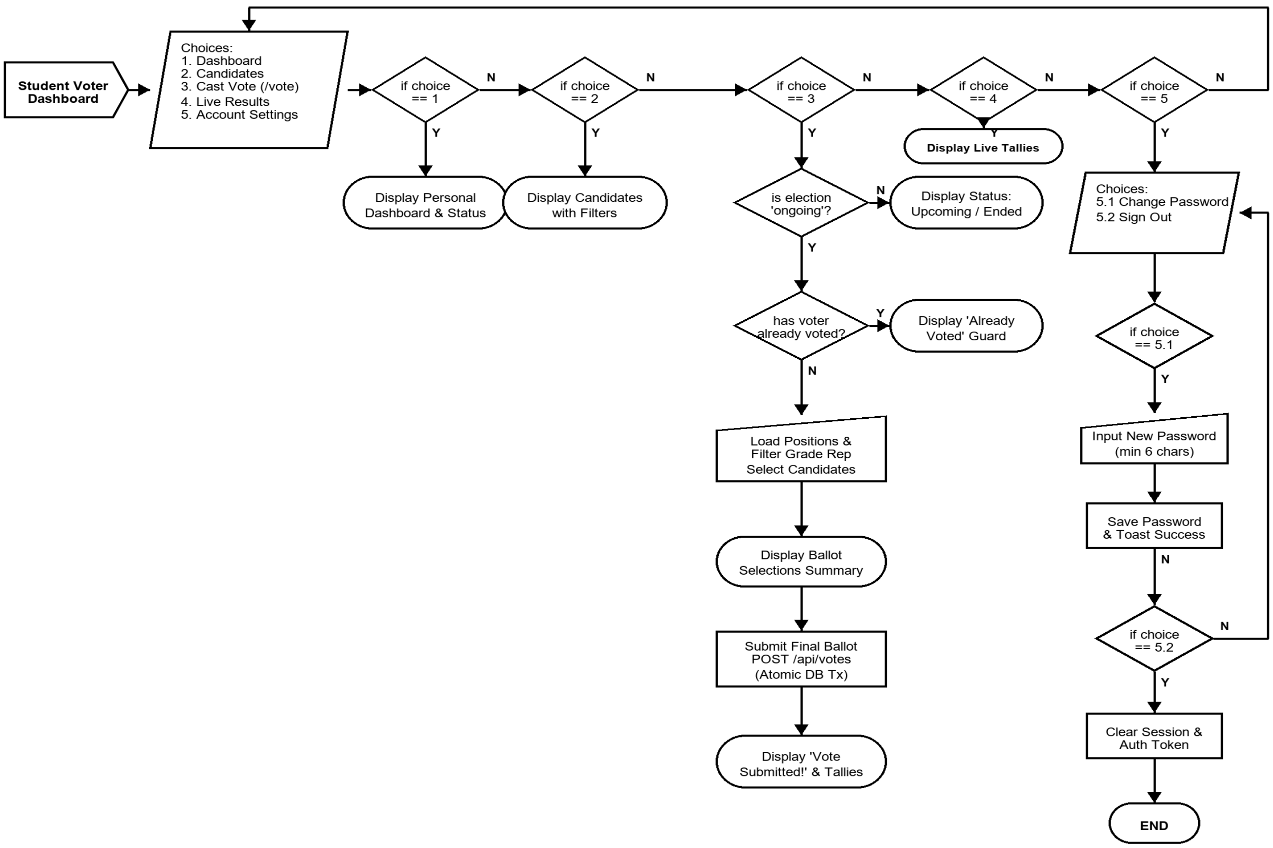

Figure 3. Student Voter Dashboard Flowchart

Figure 3 details the student voter's journey from landing on their dashboard to casting an immutable electronic ballot. The Voter Dashboard exposes choices 1 through 5:

- **Choice 1 (Dashboard):** Displays personalized voting status badge ('Voted' or 'Not Yet Voted') and election status countdown.
- **Choice 2 (Candidates Directory):** Allows exploring candidate platforms, party affiliations, and mottos.
- **Choice 3 (Cast Vote /vote):** Initiates the ballot casting engine with rigorous pre-conditions and validation.
- **Choice 4 (Live Results):** Provides voter access to real-time election statistics.
- **Choice 5 (Account Settings):** Sub-menu containing Choice 5.1 (Change Password) and Choice 5.2 (Sign Out with session invalidation).

Detailed Ballot Casting Validation Trappings (Choice 3):

• *Election State Guard:* Verifies that election_status == 'ongoing'. If 'upcoming' or 'completed', casting is blocked.
• *Duplicate Vote Prevention:* Checks if voter profile has_voted == true. If already voted, a security guard screen blocks further submission.
• *Grade Representative Constraint:* Candidates for grade-level representative positions (Grade 7 to 12) are filtered to only show candidates in voter's own grade.
• *Atomic DB Transaction:* POST /api/votes performs an ACID database transaction that inserts into votes and updates has_voted = true.

5. System Administrator Dashboard & Navigation Hub

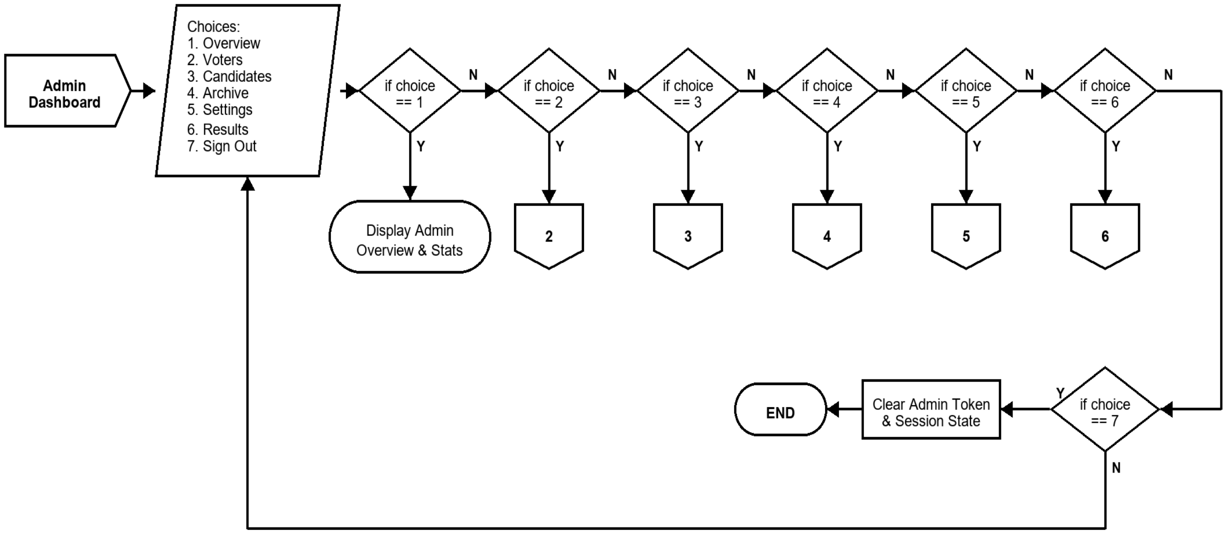

Figure 4. Admin Dashboard Flowchart

Figure 4 outlines the Master Admin Navigation Hub. Administrators authenticated with administrative role permissions gain access to centralized election governance tabs mapped directly to specialized sub-flowchart connectors:

- **Choice 1 (Overview Tab):** Renders high-level KPIs: Total Registered Voters, Total Votes Cast, Voter Turnout Percentage, and Active Candidates.
- **Choice 2 (Voters Tab -> Connector 2):** Directs to full Voter Management, single registration, CSV bulk upload, and password reset operations.
- **Choice 3 (Candidates Tab -> Connector 3):** Directs to Candidate Management, photo uploads, position assignments, and party list configuration.
- **Choice 4 (Archive Tab -> Connector 4):** Directs to Soft-Delete Archive Management for restoring or permanently deleting archived voters and candidates.
- **Choice 5 (Settings Tab -> Connector 5):** Directs to Election Lifecycle Control, scheduling, automated end crons, and historical snapshots.
- **Choice 6 (Results Tab -> Connector 6):** Directs to Real-time Results, multi-filter analytics drilldowns, and official printable tally sheets.
- **Choice 7 (Sign Out):** Prompts confirmation modal, destroys JWT token in browser storage, and terminates admin session (END).

6. Voters and Candidates Management Modules (Connectors 2 & 3)

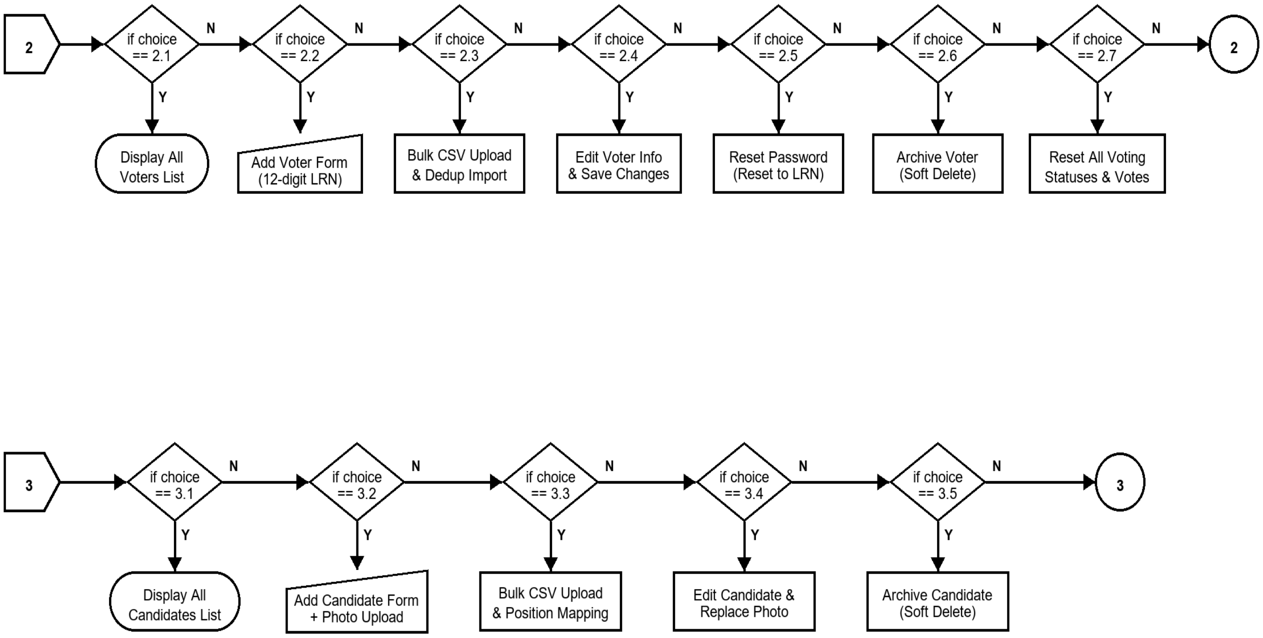

Figure 5. Voters and Candidates Management Flowchart

Figure 5 decomposes off-page Connectors 2 and 3 into detailed administrative sub-flows:

- **Connector 2 (Voters Management):** Choice 2.1 (View Voters), Choice 2.2 (Add Single Voter), Choice 2.3 (Bulk CSV Upload & Dedup), Choice 2.4 (Edit Voter Info), Choice 2.5 (Reset Password to LRN), Choice 2.6 (Soft-Delete Archive Voter), Choice 2.7 (Reset All Voting Statuses & Votes for new election).
- **Connector 3 (Candidates Management):** Choice 3.1 (View Candidates Gallery), Choice 3.2 (Add Candidate with Photo Upload to Supabase Storage), Choice 3.3 (Bulk CSV Upload & Position Mapping), Choice 3.4 (Edit Candidate & Replace Photo), Choice 3.5 (Soft-Delete Archive Candidate).

7. Archive and Recovery Management Module (Connector 4)

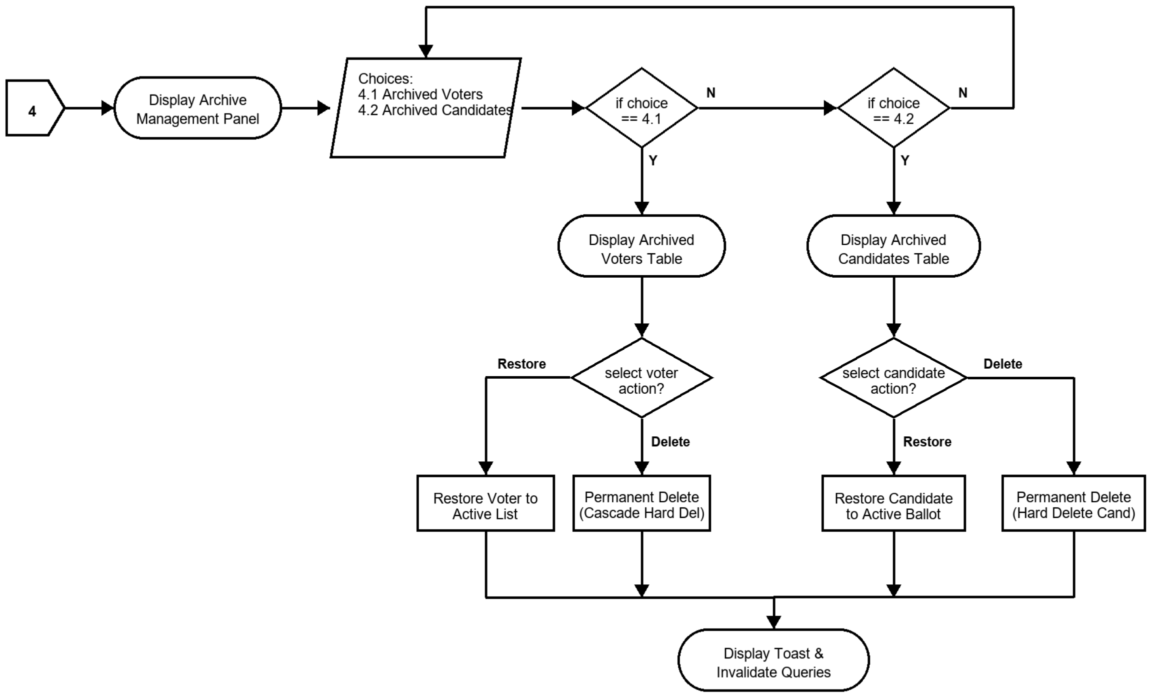

Figure 6. Archive and Recovery Management Flowchart

Figure 6 specifies the soft-delete safety architecture managed under off-page Connector 4. Accidental deletions in critical school election environments can disrupt ongoing tallies. The system segregates active and deleted entities into two sub-tabs:

- **Choice 4.1 (Archived Voters Sub-tab):** Displays list of soft-deleted voters. Admin can 'Restore' (resets archived=false, restoring voter to active roster) or 'Permanent Delete' (triggers hard SQL DELETE with foreign key cascade).
- **Choice 4.2 (Archived Candidates Sub-tab):** Displays archived candidates. Admin can 'Restore' (resumes candidacy on ballots) or 'Permanent Delete' (removes record permanently).

8. Election Settings, Schedule, and History Module (Connector 5)

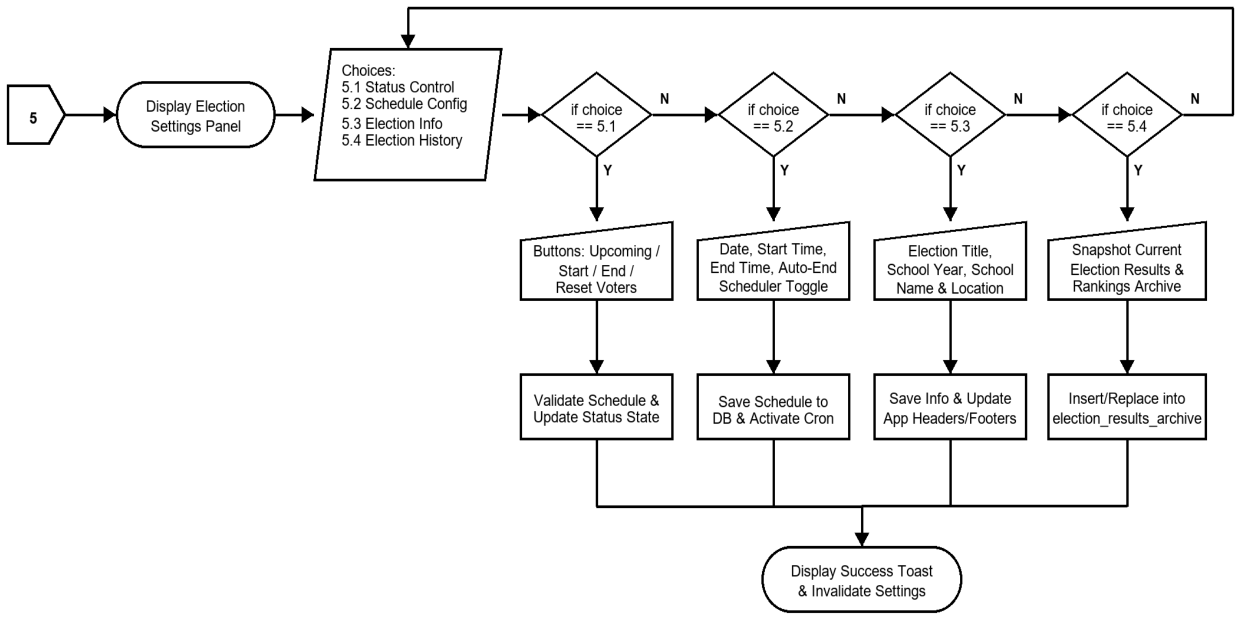

Figure 7. Election Settings, Schedule, and History Flowchart

Figure 7 details the election lifecycle state machine under off-page Connector 5, structured into four core functional areas:

- **Choice 5.1 (Election Status Control):** Controls live election states (Upcoming, Ongoing, Completed) with automatic schedule safety validation.
- **Choice 5.2 (Schedule Configuration):** Sets Election Date, Start Time (e.g. 08:00 AM), End Time (e.g. 04:00 PM), and enables automated background scheduler daemon.
- **Choice 5.3 (Election Info):** Sets School Year (e.g. 2025-2026), Election Title (e.g. SSLG General Election 2026), and School Location for system-wide branding.
- **Choice 5.4 (Election History Archive):** Captures full result snapshots into election_results_archive with candidate ranks, vote tallies, and declared winners.

9. Election Results, Analytics, and Official Tally Module (Connector 6)

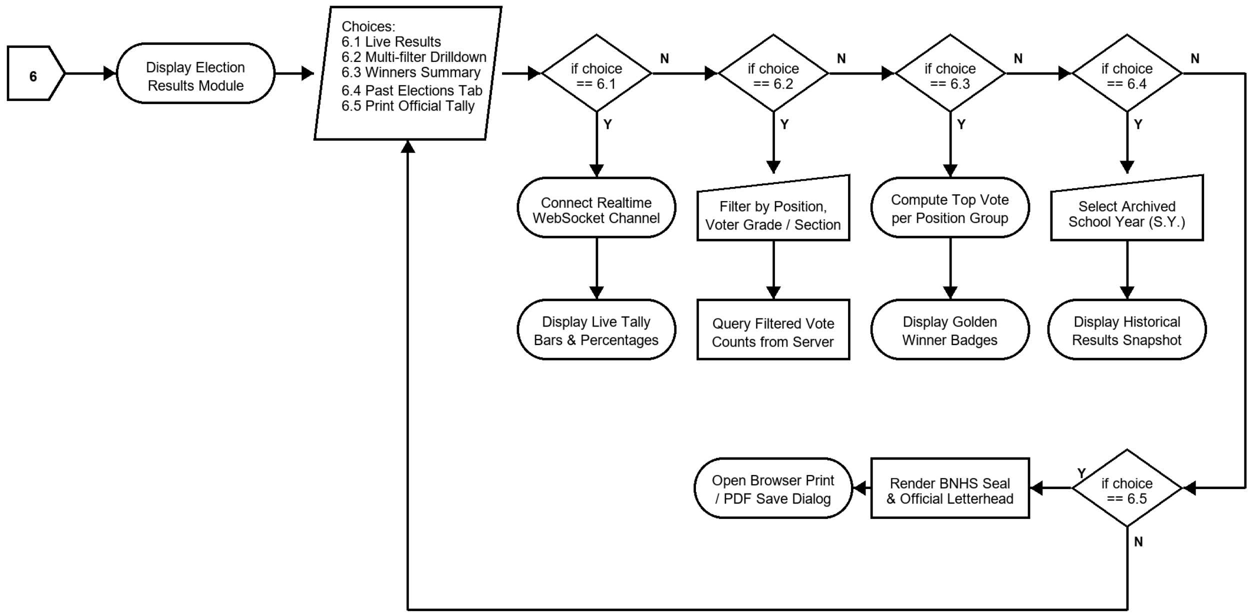

Figure 8. Election Results and Analytics Flowchart

Figure 8 outlines the election results visualization and official DepEd reporting engine under off-page Connector 6:

- **Choice 6.1 (Live Results):** Subscribes to Supabase Realtime WebSocket channel ('live-votes'), dynamically updating vote count progress bars as ballots are submitted.
- **Choice 6.2 (Multi-Filter Drilldown):** Allows filtering results by Position, Voter Grade Level (Grades 7 to 12), and Section to analyze voter participation by class.
- **Choice 6.3 (Winners Summary):** Automatically identifies and highlights the candidate with the highest votes per position with Golden Winner badges.
- **Choice 6.4 (Past Elections Tab):** Allows selecting past school years to retrieve immutable historical election outcomes.
- **Choice 6.5 (Official Print Layout):** Generates a printable DepEd-standard Results Tally Sheet complete with Batuan National High School seal, letterhead, and certification signature lines.

10. Database Schema and API Endpoint Architecture Reference

To facilitate technical development, customization, and academic defense, the table below maps the core flowchart entities to their underlying database tables and REST API endpoints:

Flowchart Component	HTTP Route	Method	Underlying Database Actions
Authentication (Fig 2)	/api/auth/login	**POST**	users, profiles (Bcrypt verify, issue JWT)
Change Password (Fig 2, 3)	/api/auth/change-password	**POST**	users (Update password_hash, clear flag)
Ballot Submit (Fig 3)	/api/votes	**POST**	votes, profiles (Atomic ACID transaction)
Voters CRUD (Fig 5)	/api/voters, /bulk-upload	**GET/POST**	users, profiles, user_roles (Manage LRNs)
Candidates CRUD (Fig 5)	/api/candidates	**GET/POST**	candidates (Photos, positions, party lists)
Archive & Restore (Fig 6)	/api/voters/:id/archive	**PUT/DEL**	profiles, candidates (Soft/Hard cascade)
Settings & Schedule (Fig 7)	/api/election-settings	**GET/PUT**	election_settings, election_results_archive

11. Human Customization and Adaptation Guide

To ensure that this documentation and its diagrams are 100% customizable by human teachers, students, and thesis panelists, multiple editing pathways are provided:

1. **Microsoft Word (.docx) Customization:** All narrative text, decision tables, step numbers, and bullet points in the accompanying .docx document are fully editable.
2. **Diagram Editing via Draw.io (diagrams.net):** The companion file Batuan_Voting_Flowcharts.drawio can be opened directly at https://app.diagrams.net to drag, drop, recolor, or modify any node.
3. **Programmatic Generation:** Python scripts in c:\batuan-voting\flowchart_build\ can be executed to regenerate 300 DPI high-resolution figures automatically.